ENVIRONMENT SCIENCE

MODULE 2

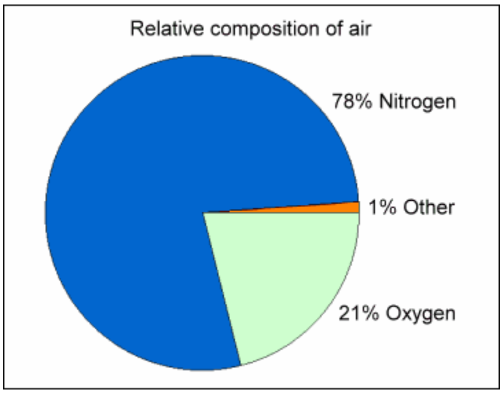
AIR

Air is one of the most important constituents of our environment. Composition of air include:

Nitrogen – 78%,

Oxygen – 21%

Other gases – 1%



AIR POLLUTION

- Air pollution is defined as "the presence of any air pollutant in the atmosphere".
- They tend to **be injurious** to human beings or other living creatures or plants or property.
- Air pollution occurs when one or several air pollutants are present in such amounts for such a long period in the outside air.
- Air pollution may be described as "the imbalance in quality of air so as to cause adverse effects on the living organisms existing on earth".

Indian Standards (IS-4167 (1996)) defines air pollution as:

"Air pollution is the presence of foreign matter in the ambient atmosphere, generally resulting from the activity of man, in sufficient concentration, present for sufficient time and under circumstances which interfere significantly with the comfort, health or welfare of person or with the full use or enjoyment of property"

AIR POLLUTANTS

Air pollutant' has been defined as 'any solid, liquid or gaseous substance present in the atmosphere in such concentration. In other way, **air pollutant** is any substance in the air that can harm human health, animals, plants, or the environment, or interfere with the comfort and enjoyment of life. These substances can be in the form of gases, particles, or biological molecules and can originate from both natural sources (like volcanic eruptions or wildfires) and human activities (like burning fossil fuels, industrial processes, or agriculture).

CLASSIFICATION OF AIR POLLUTANTS

a) Based on Origin

1. Primary Pollutants

- The pollutants that **directly cause air** pollution are known as primary pollutants
- Important primary pollutants are:
    - Carbon compounds, such as CO, CO_2, CH_4, and VOCs
    - Nitrogen compounds, such as NO, NO_2, and NH_3
    - Sulphur compounds, such as H_2S and SO_2
    - Halogen compounds, such as chlorides, fluorides, and bromides
    - Particulate Matter (PM or "aerosols"), either in solid or liquid form, categorized based on the aerodynamic diameter, such as PM10, PM2.5 etc.

a. Carbon Monoxide (CO)

- Colorless, odorless and poisonous gas
- It is produced by the **incomplete burning of carbon-based fuels** (coal, petrol, diesel and wood)
- It comes from the automobile industries, exhaust devices
- About 70% of CO emissions are from the **transport sector**
- When the air is polluted with CO, human blood is likely to be **deprived of oxygen and leads to coma and death**
- **CO** has a strong affinity towards hemoglobin of blood stream & hence inhaled can be toxic to human beings
- In mild dosages, it leads to headache

b. Oxides of Sulphur (SOx)

Sulphur dioxide (SO_2)

- SO_2 is a colorless gas
- It can be easily dissolved in water
- **SO_2** is the most common air pollutant
- SO_2 is a gas produced from **burning of coal, mainly in thermal power plants**

- Some industries such as paper mills produce SO_2
- SO_2 along with other impurities in the air can often be seen as **gray industrial smog** near industrial areas
- When water molecules in the atmosphere combine with **SO_2 and SO_3 acid rain is formed**
- It is injurious not only to men and plants, but it also attacks rapidly a few rocks such as limestone, marbles, electric contacts etc.
- This **corrodes** metals, damages buildings and monuments (Taj Mahal)
- Also sulphate particles are capable of penetrating deep into the lungs

Sulphur trioxide (SO_3)

- This is more irritant than SO2 because it combines immediately with water to form sulphuric acid

c. Oxides of Nitrogen (NOx)

Nitrogen dioxide (NO_2)

- Nitrogen dioxide (NO_2) is the most common air pollutant (which is **reddish brown in colour**)
- Major sources of NOx, are **automobile exhaust and nitric acid (HNO_3) manufacturing industries**
- NO_2 along with other impurities in the air can often be seen as a reddish-brown layer called brown photo chemical smog) over many urban areas
- When water molecules in the atmosphere combines with NO2, acid rain is formed
- This corrodes metals, damages buildings and monuments
- At higher concentration NO_2 may irritate the respiratory system **leading to cough and nasal congestions**

d. Chloro Fluoro Carbons (CFC's)

- CFC's (also known as **Freon**) are non- toxic
- They contain **Carbon, Fluorine and Chlorine atoms**
- The main CFCs are the following:
  - CFC – 11 (Trichloro Fluoro Methane $CFCl_3$)
  - CFC – 12 (Dichloro Fluoro Methane $CF2Cl_2$)
- The major uses of CFCs are as coolants in refrigerators and in air conditioners, as solvents in cleaners particularly for electronic circuit boards etc.

- CFCs are the main cause of **ozone depletion**

2. **Secondary Pollutants**

- The pollutants formed by the intermingling and reaction of primary pollutants are known as secondary pollutants
- Examples are Sulphuric acid, Ozone, Formaldehyde, Peroxy-acetyl-nitrate (PAN), Photochemical smog etc.

a. Ground-level Ozone (GL O_3)

- Ground-level Ozone is not usually emitted directly into the air, but at the ground level it is created by a chemical reaction between NOx and VOCs (Volatile Organic Compounds) in the presence of sunlight
- Ozone consists of oxygen molecules which contain three oxygen atoms
- It is not emitted directly into the air but produced in the atmosphere when oxygen combines with oxygen radical (O) in the presence of sunlight
- Ground-level Ozone may irritate the respiratory system leading to nose and throat infections and can penetrate much deep into the lungs
- In plants, it causes **premature ageing and suppressed growth**

b. Smog

- Smog is a combination of **smoke and fog** or various gases when react in the presence of sunlight
- The effects of smog on human health cause for respiratory, irritation to the eyes, diseases related to nose, throat, bronchitis, pneumonia, headache, nerves, liver, and kidneys

Two types of smog:

a. **Classical smog/London smog**: It occurs in **cool humid climate**. It was first observed in 1952 in London. It is a mixture of **smoke, fog and sulphur dioxide**. Chemically it is a reducing mixture and so it is called as reducing smog

b. **Photochemical smog/Los Angeles smog**: It occurs in **warm, dry and sunny climate**. It was first observed in 1940s in Los Angeles. The main components of the photochemical smog result from the **action of sunlight on unsaturated hydrocarbons & nitrogen oxides produced by automobiles and factories**. It has high concentration of oxidizing agents and is, therefore, called as oxidizing smog

c. **Acid rain**

- Acid rain has become one of the most important global environmental problems and poses significant adverse impact on soils, rivers, lakes, forests and monuments

- The phenomenon occurs when **SOx and NOx** from the **burning of fossil fuels** such as Petrol, Diesel, Coat etc combine with water vapour in atmosphere and **fall as rain or snow or fog**
- Natural sources like volcanoes, forest fires, etc also contribute SOx and Nox
- Increased urban and industrial activities cause air pollution resulting in the rise of concentration of SOx and NOx
- **Sulphur dioxide and NO2 combines with water vapour** in the atmosphere **produce sulphuric acid and Nitric acid respectively and results acid rain**

b) Based on Chemical Composition

1. ORGANIC

- Organic compounds containing carbon and hydrogen
- They may include carboxylic acids, ethers, esters, amines, etc.
- Natural organic of biological origin like pollen, bacteria etc
- Synthetic organic pollutant like manufactured products

2. INORGANIC

These include carbon monoxide, carbon dioxide, sulphur oxides, hydrogen chloride etc.

c) Based on state of matter

- Gaseous air pollutants: These pollutants exist in a gaseous state at normal temperature and pressure. Eg: carbon dioxide, nitrogen dioxide, sulphur oxides etc.
- Particulate air pollutants: These are not gaseous substances. Eg: suspended droplets, solid particles or mixtures of the two.

Sources of Air Pollutants

1.Natural source

- **Volcanoes**: Volcanic activity produces smoke, ash, carbon dioxide, sulfur dioxide etc.
- **Dust**: Wind-blown dust from areas with little or no vegetation such as desert areas.
- **Forest fires**: Forest fires created by natural causes, result in the formation and release of smoke, ash, dust, carbon dioxide, nitrogen oxides and other air pollutants.
- **Wetlands:** Microbial action in wetlands result in significant amounts of methane being formed and released to the atmosphere.

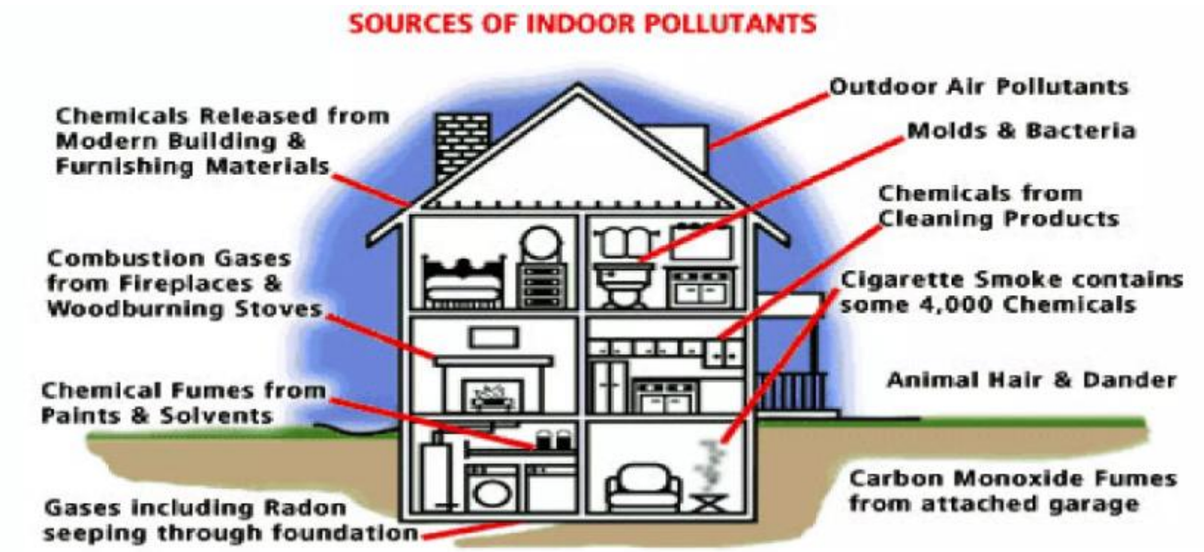
2. Anthropogenic sources

- **Stationary point sources**: A stationary point source is a single, identifiable source of air pollutant emissions. For example, the emissions from an industrial flue gas chimney / stack.
- **Mobile sources**: Mobile sources include the exhaust emissions from vehicles. A badly maintained car emits more pollution (up to 20% more volatile organic compounds and 10% more nitrogen oxides) compared to that of a well maintained one
- **Evaporative sources**: Evaporative sources are volatile liquids that, when not completely enclosed in a tank or other container, evaporate and release vapors over time. For example, liquids such as paints, solvents, pesticides, perfumes, hair sprays etc.

OUTDOOR & INDOOR AIR POLLUTANTS

INDOOR AIR POLLUTANTS

- The air pollutants which are generated from households are called indoor air pollutants.
- Indoor air pollutants are released by indoor activities such as smoking, cooking, painting, cigarette smoke, cleaning agents, mosquito repellents, etc.
- Common pollutants include particulate matter (PM), carbon monoxide, volatile organic compounds (VOCs), and biological contaminants like mold and pollen
- Poor ventilation exacerbates indoor air pollution, allowing harmful substances to accumulate
- Prolonged exposure to indoor air pollution can lead to respiratory problems, cardiovascular diseases, and other health issues, especially for vulnerable populations like children and the elderly



Controlling indoor air pollution

- Preventive measures for indoor air pollution in homes, apartments and offices involve eliminating or controlling the sources of pollution, increasing ventilation and installing air-conditioner

❖ Carefully choose building and furnishing materials (free from formaldehyde).

❖ Ensure effective ventilation in all the areas of the building (good air exchange rate)

- Install & use exhaust fans in kitchen and bathrooms
- Don't smoke at home or permit others to do so
- Use pesticides in recommended amounts with proper dilution
- Get air conditioners cleaned regularly
- Ensure cleanliness in the entire house especially in kitchen and bathrooms
- Get evaporating trays in dehumidifiers, air conditioners & refrigerators cleaned regularly
- In gas stoves, a prescient yellow tipped flame is generally an indicator of i adjustment and increased pollutant emissions. Adjust the burner so that the flame tip is blue
- Use high quality plywood in furniture's

OUTDOOR AIR POLLUTANTS

- The air pollutants which are generated outside the building are called outdoor air pollutants.
- Outdoor air pollution is primarily due to automobile exhaust, industrial emissions, mining pollutants, natural emissions from decaying matter and animals, etc.
- As per United States Environmental Protection Agency (US EPA), the major outdoor air pollutants are known as big six criteria air pollutants.
- The criteria pollutants are **carbon monoxide, lead, nitrogen dioxide, ozone, particulate matter, and sulphur dioxide.**

1. Carbon Monoxide (CO)

- Colorless, odorless and poisonous gas
- It is produced by the **incomplete burning of carbon-based fuels** (coal, petrol, diesel and wood)
- It comes from the automobile industries, exhaust devices
- About 70% of CO emissions are from the **transport sector**
- When the air is polluted with CO, human blood is likely to be **deprived of oxygen and leads to coma and death**

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- Ground-level Ozone may irritate the respiratory system leading to nose and throat infections and can penetrate much deep into the lungs.
- In plants, it causes premature ageing and suppressed growth.

3.Lead (Pb)

- Lead is a toxic material which is emitted into the atmosphere from leaded gasoline, leaded paint, lead smelters etc
- Lead is a cumulative poison and is more dangerous to children under the age of six
- Lead accumulation in the body affects the brain, causes mental retardation liver and kidney damage as well

4. Sulphur dioxide (SO_2)

- SO2 is a **colorless gas**
- It can be easily dissolved in water
- **SO_2** is the most common air pollutant
- SO_2 is a gas produced from **burning of coal, mainly in thermal power plants**
- Some industries such as paper mills produce SO_2
- SO_2 along with other impurities in the air can often be seen as **gray industrial smog** near industrial areas
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- It is injurious not only to men and plants, but it also attacks rapidly a few rocks such as limestone, marbles, electric contacts etc.
- This **corrodes** metals, damages buildings and monuments
- Also, sulphate particles are capable of penetrating deep into the lungs

5. Particulate Matter (PM)

- PM is general term used for a **mixture of solid particles and liquid droplets** found suspended in air
- They include **dust, smoke, flyash, fumes, mist, fog, aerosol** etc.
- Among these particles those which are **less than 10 µm in size remain suspended** in the atmosphere for a long period of time and are of a major concern as they can cause harmful health effects on humans
- Thus, the most harmful ones are **PM10** (coarser particles) and **PM2.5** (finer particles)

6. Nitrogen dioxide (NO_2)

- Nitrogen dioxide (NO2) is the most common air pollutant (which is **reddish brown in colour**)
- Major sources of NOx, are automobile exhaust and nitric acid (HNO3) manufacturing industries
- NO2 along with other impurities in the air can often be seen as a reddish-brown layer called brown photo chemical smog) over many urban areas
- When water molecules in the atmosphere combines with NO2, acid rain is formed
- This corrodes metals, damages buildings and monuments.
- At higher concentration NO2 may irritate the respiratory system leading to cough and nasal congestions

Impact of Air Pollutants

Air pollution is a major environmental problem that affects not only living things but also our material and property. In general, the effects of air pollution can be classified as follows:

1.Effects on human health

- The effects of air pollution on human health generally occur as a result of contact between the pollutants and the body.
- The **health depends on the concentration of contaminant present**, **time of exposure to the contaminant and the vulnerability of the person**
- Some of the common health effects are:
- ❖ Eye irritation
- ❖ Nose and throat irritation
- ❖ Increase in mortality rate

- ❖ Chronic pulmonary diseases like bronchitis and asthma [generally known as Chronic Obstructive Pulmonary Disease (COPD)]
- ❖ Carbon monoxide readily combines with hemoglobin in blood thus replacing oxygen from blood
- ❖ Carcinogenic agents cause cancer

2.Effects on plants

Some of the common effects on plants due to air pollutants are given below.

- Suppressed growth and premature ageing in plants
- Causes leaf bleaching which results in Chlorosis (Photosynthesis is affected due to loss of Chlorophyll)
- Premature falling of leaves (known as abscission)
- Causes necrosis (dead spots on the leaf structure)

3.Effects on animals & birds

Some of the common effects on animals due to air pollutants are given below:

- Affects the mucous lining of the respiratory tract
- Causes bronchitis and asthma
- Lack of appetite in pet animals
- Acid deposition cause aquatic life damage
- Migration of seasonal birds are hampered due to severe air pollution

4.Effects on materials and property

Various economic losses occur because of material damage due to air pollutants. Some of the common effects on material and property due to air pollutants are given below.

- Acid deposition can corrode metals, eat away stone on statues and monuments
- Discolor buildings, cloth fabrics

5.Effects on environment

- Severe air pollution **reduces the visibility** due to smog formation (In the presence of light, smoke from vehicles and industries react with fog to form smog)
- Air pollutants can **travel long distances** (i.e. air pollutants are carried away to distant places by the blowing wind) which results in global (transboundary") pollution.

- Some of the common examples are climate change, acid rain, global warming etc.

(Pollutants that enter the environment at one place having its effects hundreds of thousands of kilometers away from the source is known as transboundary pollution).

CLEAN AIR ACT (CAA)

The **Clean Air Act (CAA)** is a comprehensive federal law in the United States that was enacted to regulate air pollution and protect public health and the environment from harmful emissions. Initially passed in **1963** and significantly amended in **1970**, **1977**, and **1990**, the CAA provides the framework for controlling air quality and reducing pollution at a national level.

Key Objectives of the Clean Air Act

1. **Improve Air Quality**
    - Set and enforce standards to reduce air pollution.
    - Protect against harmful effects of air pollutants on human health and the environment
2. **Regulate Emissions**
    - Address emissions from both stationary sources (e.g., factories, power plants) and mobile sources (e.g., vehicles, airplanes)
3. **Protect Public Health**
    - Establish limits for pollutants that can harm health, particularly for vulnerable populations like children, the elderly, and those with preexisting health conditions
4. **Preserve the Environment**
    - Reduce pollutants that cause environmental issues such as acid rain, smog, and damage to ecosystems

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

National Ambient Air Quality Standards (NAAQS) are standards for air quality that are set by the Central Pollution Control Board (CPCB) that are applicable all over the country. NAAQS stands for **National Ambient Air Quality Standards**, which are the maximum levels of certain pollutants that are allowed in the air. The standards are set to protect public health, property, and vegetation.

The objectives of air quality standards are:

- To **indicate the levels of air quality** necessary with an adequate margin of safety to protect the public health, vegetation and property

- To assist in establishing priorities for **abatement and control of pollutant level**
- To provide uniform yardstick for **assessing air quality** at national level
- To indicate the **need and extent of monitoring programme**

National Ambient Air Quality Standards
(As of August 2011)

	Primary Standards		Secondary Standards	
Pollutant	**Level**	**Averaging Time**	**Level**	**Averaging Time**
Carbon monoxide	9 ppm (10 mg/m^3)	8-hour (1)	None	
	35 ppm (40 mg/m^3)	1-hour (1)		
Lead	0.15 µg/m^3 (2)	Rolling 3-Month Average	Same as Primary	
Nitrogen dioxide	53 ppb (3)	Annual (Arithmetic Average)	Same as Primary	
	100 ppb	1-hour (4)	None	
Particulate Matter (PM_{10})	150 µg/m^3	24-hour (5)	Same as Primary	
Particulate Matter ($PM_{2.5}$)	15.0 µg/m^3	Annual (6) (Arithmetic Average)	Same as Primary	
	35 µg/m^3	24-hour (7)	Same as Primary	
Ozone	0.075 ppm (2008 std)	8-hour (8)	Same as Primary	
	0.08 ppm (1997 std)	8-hour (9)	Same as Primary	
	0.12 ppm	1-hour (10)	Same as Primary	
Sulfur dioxide	0.03 ppm (11) (1971 std)	Annual (Arithmetic Average)	0.5 ppm	3-hour (1)
	0.14 ppm (11) (1971 std)	24-hour (1)		
	75 ppb (12)	1-hour	None	

AIR QUALITY INDEX (AQI)

- Air Quality Index (AQI) is a number used by air quality monitoring agency to characterize the quality of air at a given location.
- The main objective of AQI is to inform the public about the quality of air in their vicinity.
- The National Air Quality Index (AQI) was launched in New Delhi on September 17, 2014 under the Swachh Bharat Abhiyan
- The Central Pollution Control Board along with State Pollution Control Boards has been operating National Air Monitoring Program (NAMP) covering 240 cities of the country having more than 342 monitoring stations

- There are **six AQI categories, namely Good, Satisfactory, moderately polluted, Poor, Very Poor, and Severe**
- The proposed AQI will consider **eight pollutants (PM10, PM2.5, NO_2, SO_2, CO, O_3, Pb and NH_3)** for which short-term (up to 24-hourly averaging period) National Ambient Air Quality Standards are prescribed

AQI	**Associated Health Impacts**
Good (0–50)	Minimal impact
Satisfactory (51–100)	May cause minor breathing discomfort to sensitive people.
Moderately polluted (101–200)	May cause breathing discomfort to people with lung disease such as asthma, and discomfort to people with heart disease, children and older adults.
Poor (201–300)	May cause breathing discomfort to people on prolonged exposure, and discomfort to people with heart disease.
Very poor (301–400)	May cause respiratory illness to the people on prolonged exposure. Effect may be more pronounced in people with lung and heart diseases.
Severe (401–500)	May cause respiratory impact even on healthy people, and serious health impacts on people with lung/heart disease. The health impacts may be experienced even during light physical activity.

Description	AQI	PM10 µg/m³ 24 hr avg	PM2.5 µg/m³ 24 hr avg	CO ppm 8 hr avg	O3 ppb 8 hr avg	NO2 ppb 24 hr avg
Good + Satisfactory	0-100	0-100	0-60	0-1.7	0-50	0-43
Moderate	101-200	101-250	61-90	1.8-8.7	51-84	44-96
Poor	201-300	251-350	91-120	8.8-14.8	85-104	97-149
Very Poor	301-400	351-430	121-250	14.9-29.7	105-374	150-213
Severe	401-500	431-550	251-350	29.8-40	375-450	214-750

AQI Category, Pollutants and Health Breakpoints

AQI Category (Range)	PM_{10} (24hr)	$PM_{2.5}$ (24hr)	NO_2 (24hr)	O_3 (8hr)	CO (8hr)	SO_2 (24hr)	NH_3 (24hr)	Pb (24hr)
Good (0–50)	0–50	0–30	0–40	0–50	0–1.0	0–40	0–200	0–0.5
Satisfactory (51–100)	51–100	31–60	41–80	51–100	1.1–2.0	41–80	201–400	0.5–1.0
Moderately polluted (101–200)	101–250	61–90	81–180	101–168	2.1–10	81–380	401–800	1.1–2.0
Poor (201–300)	251–350	91–120	181–280	169–208	10–17	381–800	801–1200	2.1–3.0
Very poor (301–400)	351–430	121–250	281–400	209–748	17–34	801–1600	1200–1800	3.1–3.5
Severe (401–500)	430+	250+	400+	748+	34+	1600+	1800+	3.5+

Emission reduction strategies

Some of the control/ remedial measures to reduce air pollution are:

- Determine the priority air pollutants- based on health effect and the severity of the air quality problem
- Stacks, gets diluted and dispersed at a higher altitude itself. This results in reduced concentration of pollutants at the ground-level)
- Promotion of public transport, cycling & walking for local travel
- Stop excessive idling of vehicles
- Incorporate the control measures into a plan with implementation dates
- Improving infrastructure on public transport as a Greenhouse Gas (GHG) emission reduction measure (decreases traffic congestion and travel time which results in fuel saving)
- Involve the public (by consulting) in air quality management activities as a part of the strategy development process (This early consultation reduces later challenges and streamlines the process of implementation)
- Afforestation programme (plants absorb CO_2 during photosynthesis and releases O_2)
- Periodic air quality monitoring
- Adoption of stringent pollution control measures
- Reduction in the use of conventional fuels
- Promotion of use of renewable energy sources.
- Use of Diesel Particulate Filter (DPF) in vehicular exhaust system will not allow emission of PM 2.5 into the environment.
- The most effective method to control air pollution is to prevent the formation of air pollutants or to reduce the emission of air pollutants at the source itself

For this, the following strategies can be adopted:

1. Catalytic convertors are to be provided to the automobile exhaust to control emissions
2. Adoption of low sulphur fuels in vehicles
3. Use of Electrostatic precipitators, fabric filters and scrubbers (to remove particulate matter from waste gas streams)
4. Use of sorption technique (either adsorption of impurity onto a solid surface or absorption of the impurity in a liquid) to remove gaseous pollutants

GROUND-LEVEL OZONE

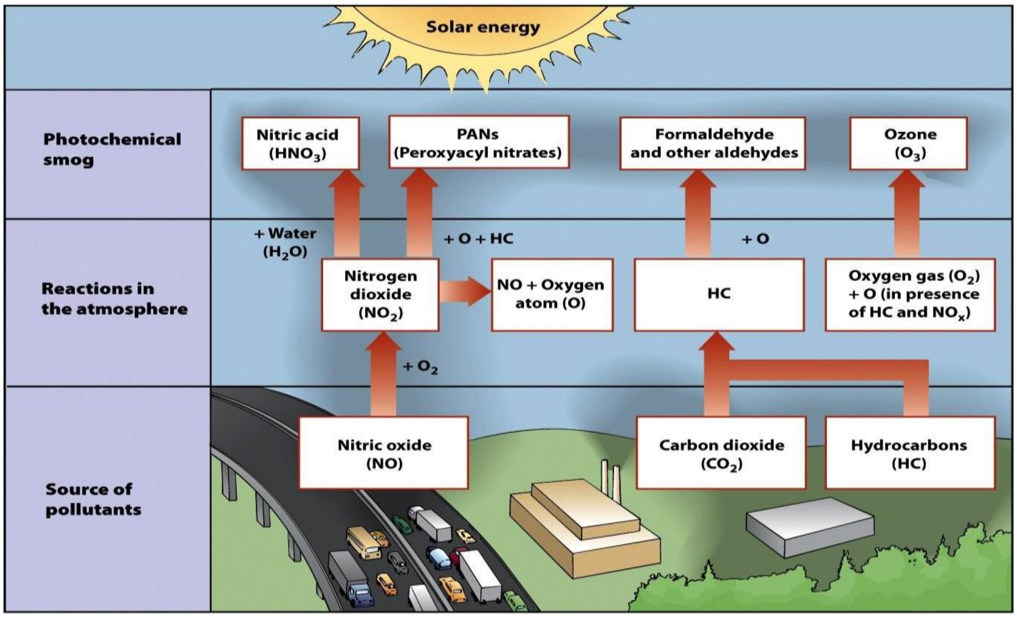
Ground-level Ozone is not usually emitted directly into the air, but at the ground level it is created by a chemical reaction between NOx and VOCs (Volatile Organic Compounds) in the presence of sunlight. Ozone consists of oxygen molecules which contain three oxygen atoms. It is not emitted directly into the air but produced in the atmosphere when oxygen combines with oxygen radical (O) in the presence of sunlight. Groundlevel Ozone may irritate the respiratory system leading to nose and throat infections and can penetrate much deep into the lungs. In plants, it causes premature ageing and suppressed growth.

Photochemical smog

Photochemical smog also known as Los Angeles smog. It occurs in **warm, dry and sunny climate**. It was first observed in 1940s in Los Angeles. The main components of the photochemical smog result from the **action of sunlight on unsaturated hydrocarbons & nitrogen oxides produced by automobiles and factories**. Photochemical smog contains pollutants such as nitrogen oxides, volatile organic compounds (VOCs), and sunlight. It has high concentration of oxidizing agents and is, therefore, called as oxidizing smog. They can cause respiratory issues and eye discomfort.

Formation of Photochemical Smog

When NOx and VOCs are released into the environment from a variety of sources, photochemical smog starts to develop. Ozone and peroxyacetyl nitrate (PAN), two secondary pollutants, are created when sunlight interacts with these pollutants and sets off a sequence of chemical reactions.



GLOBAL WARMING

- About 75% of the solar energy reaching the Earth is absorbed on the earth's surface which increases its temperature
- Global warming is the increase of average world temperatures as a result of what is known as the greenhouse effect. Certain gases in the atmosphere act like glass in a greenhouse, allowing sunlight through to heat the earth's surface but trapping the heat as it radiates back into space. As the greenhouse gases build up in the atmosphere the Earth gets hotter. This process is leading to a rapid change in climate, also known as climate change.
- Global warming is a **gradual increase in the earth's temperature** generally due to the greenhouse effect caused by increased levels of **carbon dioxide, CFCs, and other pollutants**
- Global warming, the gradual heating of Earth's surface, oceans and atmosphere, is caused by human activity, primarily the burning of fossil fuels that pump carbon dioxide (CO_2), methane and other greenhouse gases into the atmosphere.

Causes of Global warming

A. Natural causes of global warming

1.Volcanoes

Volcanoes are one of the largest natural contributors to global warming. The ash and smoke emitted during volcanic eruptions goes out into the atmosphere and affects the climate.

2.Water Vapour

Water Vapour is a kind of greenhouse gas. Due to the increase in the earth's temperature more water gets evaporated from the water bodies and stays in the atmosphere adding to global warming.

3.Melting Permafrost

Permafrost is there where glaciers are present. It is a frozen soil that has environmental gases trapped in it for several years. As the permafrost melts, it releases the gases back into the atmosphere increasing the earth's temperature.

4.Forest Blazes

Forest blazes or forest fires emit a large amount of carbon-containing smoke. These gases are released into the atmosphere and increase the earth's temperature resulting in global warming.

B. Man made causes of Global Warming

1.Deforestation

Plants are the main source of oxygen. They take in carbon dioxide and release oxygen thereby maintaining environmental balance. Forests are being depleted for many domestic and commercial purposes. This has led to an environmental imbalance, thereby giving rise to global warming.

2.Use of Vehicles

The use of vehicles, even for a very short distance results in various gaseous emissions. Vehicles burn fossil fuels which emit a large amount of carbon dioxide and other toxins into the atmosphere resulting in a temperature increase.

3.Chlorofluorocarbon

The excessive use of air conditioners and refrigerators, humans have been adding CFCs into the environment which affects the atmospheric ozone layer. The ozone layer protects the earth surface from the harmful ultraviolet rays emitted by the sun. The CFCs has led to ozone layer depletion making way for the ultraviolet rays, thereby increasing the temperature of the earth.

4.Industrial Development

With the advent of industrialization, the temperature of the earth has been increasing rapidly. The harmful emissions from the factories add to the increasing temperature of the earth.

5.Agriculture

Various farming activities produce carbon dioxide and methane gas. These add to the greenhouse gases in the atmosphere and increase the temperature of the earth.

6.Overpopulation

Increase in population means more people breathing. This leads to an increase in the level of carbon dioxide, the primary gas causing global warming, in the atmosphere.

Effects of Global warming

Global warming effects on:

1. Humans
2. Ecosystem
3. Environment
4. Economy

1. On Humans

Spread of Diseases - Global warming leads to a change in the patterns of heat and humidity. Allergies, asthma, and infectious disease outbreaks will become more common due to increased growth of pollen-producing ragweed, higher levels of air pollution, and the spread of conditions favourable to pathogens and mosquitoes.

High Mortality Rates - Due to an increase in floods, tsunamis and other natural calamities, the average death toll usually increases. Also, such events can bring about the spread of diseases that can hamper human life.

Conflicts and War - Declining amounts of quality food, water and land may be leading to an increase in global security threats, conflict and war.

2. On ecosystem:

Threats to the Ecosystem - Global warming has affected the coral reefs that can lead to the loss of plant and animal lives. Increase in global temperatures has made the fragility of coral reefs even worse.

Loss of biodiversity - A global shift in the climate leads to the loss of habitats of several plants and animals. In this case, the animals need to migrate from their natural habitat and many of them even become extinct. This is yet another major impact of global warming on biodiversity.

Increased drought - Global warming exacerbates water shortages in already water-stressed regions and is leading to an increased risk of agricultural droughts affecting crops, and ecological droughts increasing the vulnerability of ecosystems

3. On environment:

Rise in Temperature - Global warming has led to an incredible increase in earth's temperature. Since 1880, the earth's temperature has increased by 1 degree. This has resulted in an increase in the melting of glaciers, which have led to an increase in the sea level. This could have devastating effects on coastal regions.

Shrinking glaciers and rising sea level - Disappearing glaciers, early snowmelt, and severe droughts will cause more dramatic water shortages and continue to increase the risk of wildfires, rising sea levels will lead to even more coastal flooding.

Climate Change - Global warming has led to a change in climatic conditions. There are droughts at some places and floods at some. This climatic imbalance is the result of global warming.

More severe storms and floods - As temperatures rise, more moisture evaporates which worse extreme rainfall and flooding, causing more destructive storms. The frequency and extent of tropical storms is also affected by the warming ocean.

4. On economy:

Economic Consequences - Considers the cost associated with climate change rise along with the temperatures. Severe storms and floods combined with agricultural losses cause huge amount in damages, and money is needed to treat and control the spread of disease.

Extreme weather can create extreme financial setbacks. For eg. Hurricanes, floods etc.

Consumers face rising food and energy costs along with increased insurance premiums for health and home. Governments suffer the consequences of diminished tourism and industrial profits, soaring energy, food and water demands, disaster cleanup and border tensions.

OZONE LAYER DEPLETION

Ozone (O_3)

Earth's atmosphere is divided into three regions, namely troposphere, stratosphere and mesosphere. The stratosphere extends from 10 to 50 kms from the Earth's surface. This region is concentrated with slightly pungent smelling, light bluish ozone gas. The ozone gas is made up of molecules each containing three atoms of oxygen; its chemical formula is O_3. The ozone layer, in the stratosphere acts as an efficient filter for harmful solar Ultraviolet B (UV-B) rays. Ozone is produced and destroyed naturally in the atmosphere and until recently, this resulted in a well-balanced equilibrium.

Ozone Depletion Process

Ozone is highly reactive and easily broken down by man-made chlorine and bromine compounds. These compounds are found to be most responsible for most of ozone layer depletion. The ozone depletion process begins when CFCs (used in refrigerator and air conditioners) and other ozone-depleting substances (ODS) are emitted into the atmosphere. Winds efficiently mix and evenly distribute the ODS in the troposphere. These ODS compounds do not dissolve in rain, are extremely stable, and have a long-life span. After several years, they reach the stratosphere by diffusion. Strong UV light breaks apart the ODS molecules. CFCs, HCFCs, carbon tetrachloride, methyl chloroform release chlorine atoms, and halons and methyl bromide release bromine atoms. It is the chlorine and bromine atom that actually destroys ozone, not the intact ODS molecule. It is estimated that one chlorine atom can destroy from 10,000 to 100,000 ozone molecules before it is finally removed from the stratosphere.

Chemistry of Ozone Depletion

When ultraviolet light waves (UV) strike CFC ($CFCl_3$) molecules in the upper atmosphere, a carbon-chlorine bond breaks, producing a chlorine (Cl) atom. The chlorine atom then reacts with an ozone (O_3) molecule breaking it apart and so destroying the ozone. This forms an ordinary oxygen molecule (O_2) and a chlorine monoxide (ClO) molecule. Then a free oxygen atom breaks up the chlorine monoxide. The chlorine is free to repeat the process of destroying more ozone molecules. A single CFC molecule can destroy 100,000 ozone molecules.

Effects of Ozone Layer Depletion

1) Effects on Human and Animal Health: Increased penetration of solar UV-B radiation is likely to have high impact on human health with potential risks of eye diseases, skin cancer and infectious diseases.

2) Effects on Terrestrial Plants: In forests and grasslands, increased radiation is likely to change species composition thus altering the bio-diversity in different ecosystems. It could also may affect the plant community.

3) Effects on Aquatic Ecosystems: High levels of radiation exposure in tropics and subtropics may affect the distribution of Phytoplankton's, which form the foundation of aquatic food webs. It can also cause damage to early development stages of fish, shrimp, crab, amphibians and other animals, the most severe effects being decreased reproductive capacity and impaired larval development.

4) Effects on Bio-geo-chemical Cycles: Increased solar UV radiation could affect terrestrial and aquatic bio-geo-chemical cycles thus altering both sources and sinks of greenhouse and important trace gases, e.g. carbon dioxide (CO_2), carbon monoxide (CO), carbonyl sulphide (COS), etc. These changes would contribute to biosphere-atmosphere feedbacks responsible for the atmosphere build-up of these greenhouse gases.

5) Effects on Air Quality: Reduction of stratospheric ozone and increased penetration of UV-B radiation result in higher photo dissociation rates of key trace gases that control the chemical reactivity of the troposphere. This can increase both production and destruction of ozone and related oxidants such as hydrogen peroxide, which are known to have adverse effects on human health, terrestrial plants and outdoor materials.

MONTREAL PROTOCOL

- Montreal Protocol is an international treaty signed by 46 countries in September 16, 1987 to limit the production of substances harmful to the stratospheric ozone layer

- The Protocol sets out a mandatory timetable for the phase out of **Ozone Depleting Substances (ODS such as CFC, halons, and HCFCs).**
- The Montreal Protocol on Substances that Deplete the Ozone Layer is an important Multilateral Agreement regulating the production, consumption, and emissions of ozone-depleting substances (ODSs).
- It is an important part of international environmental conventions and protocols.
- The Protocol was signed in 1987 and entered into force in January 1989. The protocol gives provisions to reduce the production and consumption of ODSs to protect the ozone layer
- The Montreal Protocol **controls both consumption and production of ODS** in order to protect the interests of producers and importers, who otherwise would have had to sustain high price inflation or overproduction during the phase out period of the targeted gases
- The compounds are divided into groups: Group I (certain CFCs) and Group II (specific halons)
- The protocol also distinguishes between two groups of countries, the more developed with relatively high levels of consumption of the contoured ozone depleting substances and the developing countries with relatively low levels of consumption

NOISE

- "The word noise comes from the Latin word 'noxia' meaning "injury" or "hurt"
- Noise is an unwanted, unpleasant and annoying sound caused by vibration of the matter
- Vibrations impinge on the ear drum of a human or animal and setup a nervous disturbance, which we call sound
- **When the effects of sound are undesirable that it may be termed as Noise**
- The audible sound range for humans is typically between **20 and 20,000 Hertz (Hz)**
- The unit of sound is the decibel (dB)

Sources of noise

Noise is an unwanted sound and noise pollution occurs through different sources:

1. Vehicles produce noise that leads to noise pollution
2. Automobile industry is another source of noise pollution
3. Noise pollution is very common in industrial areas where machines are working for factories making more noise

(1) Stationary sources

a) Industrial noise: The main categories of industrial activity that are particularly relevant to the study of noise are the following: Product fabrication, Product assembly, Power generation by means of generators, Combusting process in furnaces (burning of gases)

b) Noise from construction works: Construction noise, a major source of noise pollution is emitted by construction equipment. The sources of noise are dozers excavators, front end loaders, soil compactors, cranes, air compressors, concrete vibrators, riveting steel structure during the casting, dismantling of construction materials etc.

c) Noise from other sources: These include sources such as sirens, barking dogs, ambulances, Police vehicles, Fire engines etc.

(2) Mobile sources

a) Road traffic: Of all sources of noise pollution, road traffic is the most prevalent and perhaps the most source of noise pollution. More people are exposed to noise from motor vehicles and the noise depends on various factors such as Road location, Road design, Vehicle standards, Driver behaviours, horns, traffic density.

b) Railway traffic: Noise from railway traffic is not serious nuisance as compared to the road traffic noise. The level of noise associated with rail traffic is related to the type of engine, the speed of the train, track type and condition. The majority of noise emitted by trains is produced by the engine or by the interaction of wheels with the tracks, horns, warning signals at crossings etc.

c) Air traffic: The noise of air craft is different from that of road traffic in the sense it is intermittent. Noise is maximum during take-off and landing. Noise made by jet planes is more disturbance than that of propeller driven air craft. Supersonic air craft produce noise at high levels due to its intensity

Types of noise

1.Continuous Noise – It is an uninterrupted sound level that varies less than 5 dB during entire period of observation. Example: running fan.

2. Intermittent Noise – It is a noise which continues for more than 1 second and is then interrupted for more than 1 second. Example: drilling machine used by a dentist.

3. Impulse Noise – It is characterized by a change of sound pressure of at least 40 dB within 0.5 second with a duration of less than one second. Example: firing of a weapon.

4. Traffic Noise

The main source of noise pollution is in urban areas. It includes air, road, rail, sea-shore & inland water traffic. Amount and type of noise produced by traffic is largely dependent upon type of traffic. As a rule, larger & heavier vehicles emit more noise than smaller & lighter vehicles (Exceptions would include helicopters and 2 & 3-wheeled road vehicles). Noise of road vehicles is mainly generated from engine & from frictional contact between vehicle & the ground and air. Railway noise depends on the speed of the train, the type of engine, wagons, and rails and their foundations, as well as the roughness of wheels and rails. Aircraft take-offs are known to produce intense noise, including vibration and rattle

5. Industrial noise

Noise is the essential by-product of industry. Intensity and nature depend on type of industry. Causes both indoor & outdoor noise pollution. Industrial noises are usually produced by rotating, reciprocating or any other types of machinery, or by high pressure high velocity gases, liquids or vapour (for example, fans, steam pressure relief valves) involved in the industrial processes. The usual noise level of the industries is of the order of 60 to 95 decibels. Machinery should preferably be silenced at the source.

6. Domestic Noise

In residential areas, noise may come from **mechanical devices** (e.g. heat pumps, ventilation systems and traffic), as well as **voices, music** and other kinds of sounds generated by **neighbours** (e.g. lawn mowers, vacuum cleaners and other household equipment, music reproduction and noisy parties). Includes **movement of utensils**, **cutting** and **peeling of fruits/vegetables** etc. Activities like **plumbing, boilers, generators, air conditioners, fans, and vacuum cleaners** add to the existing noise pollution

7. Construction noise

Construction of building highways, and streets cause a lot of noise due to the usage of **air compressors, bulldozers, loaders, dump trucks, and pavement breakers**. A variety of sounds come from **cranes, cement mixers, welding, hammering, boring and other work processes**. Construction equipment is often poorly silenced and maintained, and building operations are sometimes carried out without considering the environmental noise consequences.

8. Noise from other miscellaneous sources

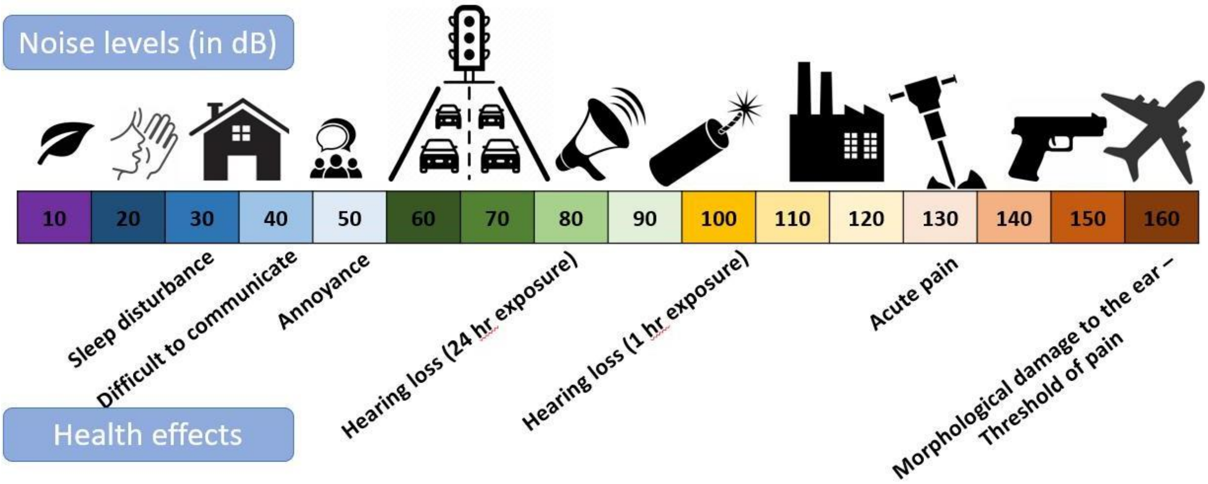
Several other human activities, such as blaring of loud-speakers and sirens, shouting of hawkers, playing of children, general life and activity, ringing of temple and church bells, etc. do produce noises of different levels, tones and spectra.

Levels of Noise

- Faintest sound by human ear - 20 µPa
- Loudest sound by human ear – 200 Pa
- Sound pressure represented on a scale based on the log of the ratio of the measured sound pressure and a reference standard pressure
- Measurements on this scale are called levels
- Since the unit of sound level turns out to be larger unit, a smaller unit of decibels (dB) is used:

$$1\text{dB} = (1/10)\text{ B}$$

B – Bel



Effects of noise

- Exposure to loud noise is one of the major causes, apart from ageing
- Hearing loss is the second most common disability among humans
- Prevalence of hearing loss increases with age

Effects of noise are categorized into two:

a) <u>Auditory Health Effects</u>

- **Acoustic trauma** - Sudden hearing damage caused by short burst of extremely loud noise such as a gun shot
- **Tinnitus** - Characterized by a ringing/buzzing sound in the ears

- **Temporary hearing loss** - Caused due to exposure to very loud noises. Gradual recovery when the affected person spends time in a quiet place
- **Noise-induced hearing loss** - Caused due to a one-time exposure to extremely loud noise or steady long-term exposure to noise levels exceeding 75-85 dB. Characterized by loss of auditory sensory cells in the cochlea. Hearing loss is permanent because the cochlear cells cannot regenerate. Prevention of noise-induced hearing loss is the only option to preserve hearing.

b) Non-auditory Health Effects

- **Annoyance -** A general feeling of displeasure or adverse reaction triggered by the noise. The degree of annoyance depends on the characteristics of noise & personal/situational factors
- **Sleep disturbance -** Night time noise can cause frequent awakenings, reduced sleep quality, increased drowsiness, tiredness, decreased performance, & insomnia
- **Impaired cognitive functioning -** Leads to difficulty in concentrating, poor reading and writing skills, difficulty in remembering, affects speech intelligibility, poor overall performance in school
- **Cardiovascular disease -** Changes in blood pressure and increased risk of various types of heart disease
- **Mental illness -** Noise may well accelerate and intensify the development of latent mental disorder

Noise pollution control measures

There are four fundamental ways in which noise can be controlled:

- Reduce noise at the source
- Block the path of noise
- Increase the path length
- Protect the recipient

In general, the best control method is to reduce noise levels at the source. Noise pollution could be controlled by either reducing the noise at the source or by preventing its transmission. The first step in the prevention of noise pollution is to control the noise at source itself. The first step in the prevention of noise pollution is to control the noise at source itself

- For eg: Lubrication of machines reduces the noise produced, Tightening the loose nuts, Reducing the vibrations produced by machines etc.

Failing to control the noise at its source, the second step is to prevent its transmission

- For eg: keeping the noise machine covered in an enclosure so that the sound does not escape and reach the receivers

If the noise levels are not able to bring down to the desired levels in some cases, the only alternative is to follow:

- Avoiding horns except in emergency situations
- Sound proof or eco-generators and turning down the volume of stereos
- Conducting the awareness programs

a) Control at Source

1.Reducing the noise levels from domestic sectors

- By their selective and judicious operation
- By usage of carpets or any absorbing materials inside houses

2.Maintenance of automobiles

- Regular servicing and tuning of vehicles will reduce the noise levels
- Fixing of silencers to automobiles & two wheelers, etc.

3.Control over vibrations

- The vibrations of materials may be controlled using proper foundations, rubber padding etc. to reduce the noise levels caused by vibrations

4.Prohibition on usage of loudspeakers

- By not permitting the usage of loudspeakers in the habitant zones except for important meetings / functions

5. **Selection of machinery**

- Optimum selection of machinery tools or equipment reduces excess noise levels.

6.Maintenance of machines

- Proper lubrication and maintenance of machines, vehicles etc. will reduce noise levels

7.Use electric vehicles

- At low speeds produce very low levels of noise
- Silent and unnoticeable
- Difference between noise of electric vehicle & IC engine vehicle > 6 dBA at 10 km/h

8.Smart traffic management

- Centrally controlled traffic signals and sensors regulate the flow of traffic through the city in compliance with the current state on the roads in the city

9.Quiet road surfaces

- Thin surface layers, two-layer porous asphalt & cast asphalt can be used as road surface layers to reduce the noise levels by 4 dB, 7 dB & 1.5 dB respectively

10. **Production of quiet car tires**

- Smaller tire represents a smaller surface that adheres to road - produces less noise
- Softer types of rubber - make less noise.

11. **Efficient traffic management**

- Reducing traffic density
- Reducing the percentage of heavy goods vehicles

12. **Change in driving behavior**

- Speed reduction / traffic calming measures
- Promoting environmentally friendly means of transport, e.g. walking and cycling

b)Control in Transmission Path

1. Installation of barriers - Installation of barriers between noise source and receiver can attenuate the noise levels

2. Green belt development

- Green belt development can attenuate the sound levels
- The degree of attenuation varies with species of greenbelt

3.Design of building

- The design of the building incorporating the use of suitable noise absorbing material for wall / door / window/ceiling will reduce the noise levels

4.Installation of panels or enclosures

- A sound source may be enclosed within a panelled structure such as room as a means of reducing the noise levels at the receiver

c) Control at Receiver

1.Using hearing protection equipment

- Equipment like earmuffs & ear plugs are commonly used devices for hearing protection
- Average noise attenuation up to 32 dB can be achieved using earmuffs

2.Exposure reduction

- Schedule of workers should be planned in such a way that they should not be overexposed to the high noise levels

3.Job rotation

- By rotating the job between the workers working at a particular noise source or isolating a person, the adverse impacts can be reduced

Quantification of noise pollution

Leq

- Stands for **"equivalent noise level" or "equivalent continuous sound level"**.
- It's the average sound level over a given period of time
- It is calculated as the constant noise level that would produce the same total sound energy as the fluctuating sounds
- Leq is a **fundamental measurement parameter** used in environmental noise standards and other regulations.
- Leq is a better indicator of potential hearing damage or the likelihood of complaints than a simple instantaneous sound level
- It's also easier to read accurately than a simple instantaneous sound level.
- A sound level meter that measures Leq is called an integrating sound level meter

LAeq

- Stands for "A-weighted, equivalent continuous sound level"
- It's the Leq measurement that's frequency weighted
- LAeq is widely used in environmental noise assessments, workplace noise regulation and urban planning to understand long-term exposure to varying noise levels

NOISE REGULATION RULES IN INDIA

<u>THE NOISE POLLUTION (REGULATION AND CONTROL) RULES, 2000</u>

The increasing ambient noise levels in public places from various sources, inter-alia, industrial activity, construction activity, generator sets, loud speakers, public address terms, music systems, vehicular horns and other mechanical devices have mysterious effects on human health and the psychological well being of the people; it is considered necessary to regulate and control noise producing and venerating sources with the objective of maintaining the ambient air quality standards in respect of noise.

Ambient Air Quality standards in respect of Noise

Area Code	Category of Area/Zone	Limits in dB (A) Leq*	
		Day Time	Night Time
A	Industrial Area	75	70
B	Commercial Area	65	55
C	Residential Area	55	45
D	Silence Zone	50	40

- Day time shall mean from 6.00 am. to 10.00 pm
- Night time shall mean from 10.00 pm. to 6.00 am
- Silence zone is defined as an area comprising not less than 100 meters around hospitals, educational institutions and courts
- The silence zones are zones, which are declared as such by the competent authority
- Mixed categories of areas may be declared as one of the four mentioned categories by the competent authority